
Problem 8.30. Show that E_{DFA} is NL-complete.

Proof. To show that E_{DFA} is NL-complete, we must show that it is in NL and that all NL problems are log space reducible to it. To show $E_{\text{DFA}} \in \text{NL}$, we show that $\overline{E_{\text{DFA}}} \in \text{NL}$. We give an algorithm M that decides $\overline{E_{\text{DFA}}}$ in non-deterministic log space.

$M =$ "On input $\langle D \rangle$, where $D = (Q, \Sigma, \delta, q_0, F)$ is a DFA and every state $q_i \in Q$ is a binary integer greater than 0:

1. Let i, k, n and q be binary integers on work tape.
2. Let $n = |Q|$.
3. Non-deterministically, repeat for each $k = 0, 1, \ldots, n$:
4. Let $q = q_0$ and $i = 0$.
5. If $i < k$, then:
6. Non-deterministically, repeat for each $a \in \Sigma$:
7. Let $q = \delta(q, a)$ and $i = i + 1$.
8. Go to step 5.
9. If $q \in F$, then *accept*. Otherwise, *reject*."

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